Worksheet 6.1 — Antiderivatives and Indefinite Integrals

Mathematics: Analysis & Approaches SL (IB) · Unit 6: Integral Calculus

The Big Ideas

Integration reverses differentiation; every indefinite integral carries an unknown constant C.

- $\displaystyle\int x^n\,dx = \frac{x^{n+1}}{n+1} + C.$
- $\displaystyle\int (ax+b)^n\,dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C.$

Worked Examples

Example 1 — Power rule

Find $\displaystyle\int (12x^3 - 4x + 7)\,dx$.

Solution. $3x^4 - 2x^2 + 7x + C$.

Conclusion: $3x^4 - 2x^2 + 7x + C$.

Example 2 — Finding C

Given $f'(x) = 4x - 3$, $f(2) = 5$, find $f(x)$.

Solution. $f(x) = 2x^2 - 3x + C$; $5 = 8 - 6 + C \Rightarrow C = 3$.

Conclusion: $f(x) = 2x^2 - 3x + 3$.

Example 3 — Linear substitution

Find $\displaystyle\int (4x-1)^5\,dx$.

Solution. Divide by $a = 4$: $\dfrac{(4x-1)^6}{24}$.

Conclusion: $\dfrac{(4x-1)^6}{24} + C$.

Example 4 — From acceleration to position

$a(t) = 8t - 6$, $v(0) = 2$, $s(0) = -3$. Find $s(t)$.

Solution. $v(t) = 4t^2 - 6t + 2$ (using $v(0) = 2$). $s(t) = \frac{4}{3}t^3 - 3t^2 + 2t - 3$ (using $s(0) = -3$).

Conclusion: $s(t) = \frac{4}{3}t^3 - 3t^2 + 2t - 3$.

Example 5 — Rewriting a root before integrating

Find $\int \sqrt{x}\,dx$ by rewriting as a fractional power.

Solution. $\int x^{1/2}\,dx = \dfrac{x^{3/2}}{3/2}$.

Conclusion: $\dfrac{2}{3}x^{3/2} + C$.

Example 6 — Splitting a fraction with two terms

Find $\int \dfrac{6x^2 + 4x}{2x}\,dx$.

Solution. Split termwise: $\dfrac{6x^2}{2x} + \dfrac{4x}{2x} = 3x + 2$.

Conclusion: $\dfrac{3x^2}{2} + 2x + C$.

Example 7 — Finding a curve given its slope and a point

A curve has $\dfrac{dy}{dx} = 6x^2 - 4x + 1$ and passes through $(1, 3)$. Find y.

Solution. $y = 2x^3 - 2x^2 + x + C$; substitute $(1, 3)$: $3 = 2 - 2 + 1 + C \Rightarrow C = 2$.

Conclusion: $y = 2x^3 - 2x^2 + x + 2$.

Example 8 — Two-stage integration with two conditions

Given $f''(x) = 12x$, $f'(0) = 3$, $f(0) = -1$, find $f(x)$.

Solution. $f'(x) = 6x^2 + 3$ (using $f'(0) = 3$). $f(x) = 2x^3 + 3x - 1$ (using $f(0) = -1$).

Conclusion: $f(x) = 2x^3 + 3x - 1$.

Example 9 — Application — reconstructing distance from acceleration

A car accelerates according to $a(t) = 2$ m/s^2 starting from rest ($v(0) = 0$) at position $s(0) = 0$. Find its position after $t = 6$ s.

Solution. $v(t) = 2t$; $s(t) = t^2$.

Conclusion: $s(6) = 36$ m.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Find $\int (15x^4 - 8x + 2)\,dx$.

Question 2

Given $f'(x) = 6x^2 + 1$, $f(1) = 4$, find $f(x)$.

Question 3

Find $\int (2x + 7)^4\,dx$.

Question 4

Find $\int \dfrac{9x^3 - 6x}{3x}\,dx$.

Question 5 — Challenge

A particle has $a(t) = 6t - 4$, $v(0) = 0$, $s(0) = 2$; find $s(t)$.

Question 6 — Challenge

Find $\int (3x-2)^3\,dx$, and check your answer by differentiating it.

Question 7

Find $\int \sqrt[3]{x}\,dx$ (rewrite as a fractional power first).

Question 8

Find $\int \dfrac{10x^3-15x}{5x}\,dx$.

Question 9

A curve has $\dfrac{dy}{dx} = 3x^2 - 6x + 2$ and passes through $(2, 1)$. Find y.

Question 10 — Challenge

Given $f''(x) = 6x - 4$, $f'(1) = 1$, $f(0) = 2$, find $f(x)$.

Question 11

A rocket accelerates at $a(t) = 4$ m/s^2 from rest at $s(0) = 0$; find its position after $t = 5$ s.

Question 12

Find $\int (x+3)(x-2)\,dx$ by expanding first.

Question 13 — Challenge

Find $\int (5x+2)^{-2}\,dx$ (treat the negative exponent like any other power in the linear-substitution pattern).

Question Recap & Answer Key

Question Recap

1. Find $\int (15x^4 - 8x + 2)\,dx$.
2. Given $f'(x) = 6x^2 + 1$, $f(1) = 4$, find $f(x)$.
3. Find $\int (2x + 7)^4\,dx$.
4. Find $\int \dfrac{9x^3 - 6x}{3x}\,dx$.
5. A particle has $a(t) = 6t - 4$, $v(0) = 0$, $s(0) = 2$; find $s(t)$.
6. Find $\int (3x - 2)^3\,dx$, and check your answer by differentiating it.
7. Find $\int \sqrt[3]{x}\,dx$ (rewrite as a fractional power first).
8. Find $\int \dfrac{10x^3 - 15x}{5x}\,dx$.
9. A curve has $\dfrac{dy}{dx} = 3x^2 - 6x + 2$ and passes through $(2, 1)$. Find y.
10. Given $f''(x) = 6x - 4$, $f'(1) = 1$, $f(0) = 2$, find $f(x)$.
11. A rocket accelerates at $a(t) = 4$ m/s^2 from rest at $s(0) = 0$; find its position after $t = 5$ s.
12. Find $\int (x + 3)(x - 2)\,dx$ by expanding first.
13. Find $\int (5x + 2)^{-2}\,dx$ (treat the negative exponent like any other power in the linear-substitution pattern).

✓ Answer Key

1. $3x^5 - 4x^2 + 2x + C$
2. $f(x) = 2x^3 + x + C$; $4 = 2 + 1 + C \Rightarrow C = 1$: $f(x) = 2x^3 + x + 1$
3. $\dfrac{(2x + 7)^5}{10} + C$
4. simplify to $3x^2 - 2$: $x^3 - 2x + C$
5. $v(t) = 3t^2 - 4t$; $s(t) = t^3 - 2t^2 + 2$
6. $\dfrac{(3x - 2)^4}{12} + C$; differentiating gives $\dfrac{4(3x - 2)^3 \cdot 3}{12} = (3x - 2)^3$, matching the original integrand
7. $\int x^{1/3} dx = \dfrac{x^{4/3}}{4/3} = \dfrac{3}{4}x^{4/3} + C$
8. simplify to $2x^2 - 3$: $\dfrac{2x^3}{3} - 3x + C$
9. $y = x^3 - 3x^2 + 2x + C$; $1 = 8 - 12 + 4 + C \Rightarrow C = 1$: $y = x^3 - 3x^2 + 2x + 1$
10. $f'(x) = 3x^2 - 4x + C_1$; $1 = 3 - 4 + C_1 \Rightarrow C_1 = 2$; $f(x) = x^3 - 2x^2 + 2x + C_2$; $2 = 0 + C_2 \Rightarrow C_2 = 2$: $f(x) = x^3 - 2x^2 + 2x + 2$
11. $v(t) = 4t$; $s(t) = 2t^2$; $s(5) = 50$ m
12. expand to $x^2 + x - 6$: $\dfrac{x^3}{3} + \dfrac{x^2}{2} - 6x + C$
13. $\int (5x+2)^{-2} dx = \dfrac{(5x + 2)^{-1}}{5(-1)} + C = -\dfrac{1}{5(5x + 2)} + C$